

MorphNN: Auditable Open-Set Viral Exanthem Triage Using Morphology-Informed Frozen Feature Fusion

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Abstract:

Viral exanthem triage is a public-health problem as much as an image-classification problem. In resource-constrained settings, early differentiation of visually similar rashes can support isolation, referral, and confirmatory testing, yet automated systems are often optimized as closed-set classifiers without equal attention to computational feasibility, auditability, or uncertainty under atypical presentations. This study investigates whether explicit lesion morphology, when fused with compact frozen texture representations, can provide a more practical alternative to fully fine-tuned convolutional neural networks for Varicella–Measles differentiation. We propose MorphNN, a morphology-informed frozen feature fusion framework that represents each image using an 18-dimensional decision vector. Eight deterministic descriptors quantify lesion burden, area distribution, circularity, spatial dispersion, confluence, and localized colour characteristics, while ten PCA-compressed texture components are extracted from a frozen MobileNetV2 backbone. The fused representation is classified using a Random Forest ensemble, allowing feature-level audit of the final decision space without end-to-end gradient adaptation. Across a four-phase evaluation comparing morphology-only classifiers, frozen transfer-learning baselines, fine-tuned CNNs, and hybrid fusion, MorphNN achieved a holdout macro F1-score of 0.957, closely approaching the best fine-tuned EfficientNet-B0 baseline of 0.963. The optimal MobileNetV2 + Random Forest configuration achieved 0.953 ± 0.007 macro F1, with 14.2 ± 0.6 s training time, 9.8 ± 0.4 ms/image CPU inference latency, and a 16.1 MB deployable pipeline. Feature audit showed that the classifier used both latent texture information and clinically meaningful morphology, particularly confluence density, suggesting that frozen texture embeddings mitigate segmentation fragility without replacing explicit lesion descriptors.

[NEW RESULTS - Add open-set/reject-option performance against Monkeypox, HFMD, Cowpox, and Healthy classes as unknown presentations.]

[NEW RESULTS - Add calibration and uncertainty findings, including ECE, Brier score, reliability behaviour, or threshold-based referral performance.]

These findings position MorphNN as a compact, auditable, and resource-conscious framework for viral exanthem triage rather than an accuracy-only classifier. Open-set validation and calibration remain necessary before clinical deployment.

Abstract:

Automated rash recognition has limited clinical value if it performs well only under closed-set assumptions while remaining difficult to inspect, expensive to train, or unsuitable for resource-constrained settings. Varicella and measles are clinically consequential viral exanthems with overlapping visual manifestations in routine photographs, where lesion stage, image quality, illumination, and rash morphology can obscure class boundaries. This study investigates whether explicit lesion structure, supported by compact frozen texture representations, can approximate the performance of fine-tuned convolutional neural networks while offering a more auditable and deployable diagnostic triage pathway. We propose MorphNN, a morphology-informed frozen feature fusion framework for Varicella–Measles differentiation. Each image is represented by an 18-dimensional standardized vector comprising eight deterministic morphology descriptors and ten PCA-compressed texture components extracted from a frozen MobileNetV2 backbone. The morphology stream captures lesion burden, area distribution, circularity, spatial dispersion, confluence, and localized colour characteristics, while the frozen texture stream provides complementary visual information without end-to-end gradient adaptation. The fused vector is classified using a Random Forest ensemble, enabling feature-level audit of the final decision space. Across a four-phase evaluation comparing morphology-only classifiers, frozen transfer-learning baselines, fine-tuned CNNs, and hybrid fusion, MorphNN achieved a holdout macro F1-score of 0.957, closely approaching the best fine-tuned EfficientNet-B0 baseline of 0.963. The optimal MobileNetV2 + Random Forest configuration achieved 0.953 ± 0.007 macro F1, with 14.2 ± 0.6 s training time, 9.8 ± 0.4 ms/image CPU inference latency, and a 16.1 MB deployable pipeline. Feature audit indicated that the fused classifier retained clinically meaningful morphology, particularly confluence density, while using frozen texture components to reduce segmentation-related fragility.

[NEW RESULTS - Add open-set/reject-option performance using Monkeypox, HFMD, Cowpox, and Healthy classes as unknown presentations; report AUROC, AUPR, false-accept rate, or rejection accuracy.]

[NEW RESULTS - Add calibration and uncertainty findings, such as ECE, Brier score, reliability-curve behaviour, or threshold-based referral performance.]

These findings position MorphNN as a compact, auditable, and resource-conscious framework for viral exanthem triage rather than an accuracy-only classifier. Open-set validation and calibration remain necessary before any clinical deployment claim.

1. Introduction:

Viral exanthems remain a persistent diagnostic and public-health challenge, particularly in settings where specialist dermatology, laboratory confirmation, and high-performance computing infrastructure are not readily available. Measles is a highly contagious vaccine-preventable disease associated with fever, respiratory symptoms, conjunctivitis, and a maculopapular rash that commonly spreads from the face and upper neck to the trunk and extremities [1], [10], [NEW REF]. Varicella, in contrast, is typically characterized by pruritic lesions that may progress from macules to papules, vesicles, and crusts, often with lesions at different stages of evolution [NEW REF]. Although these diseases differ in virology, epidemiology, management, and infection-control implications, their visual presentation can overlap in routine clinical photographs, especially when rash stage, skin tone, illumination, camera quality, and image framing are variable. In such contexts, image-based decision support is not simply an accuracy problem; it is a triage problem in which misclassification may influence isolation decisions, referral pathways, and confirmatory testing.

Recent advances in deep learning have substantially improved automated skin-lesion and viral-rash classification. Transfer-learning models based on convolutional neural networks, including ResNet, EfficientNet, MobileNet, VGG, and transformer-hybrid architectures, have been applied to dermatological image benchmarks and viral lesion datasets with promising performance [4], [5], [18]–[21]. The Multi-Class Viral Skin Lesion Dataset (MCVSLD) has further enabled reproducible experimentation across multiple viral and non-viral skin categories, including Chickenpox, Measles, Monkeypox, HFMD, Cowpox, and Healthy images [13]. However, much of the existing work frames the task primarily as closed-set classification, where every test image is forced into one of the known training categories. This assumption is convenient experimentally but clinically restrictive: in real-world triage, an image may reflect an atypical presentation, a different viral exanthem, a non-infectious rash, poor image quality, or a condition outside the model’s validated diagnostic scope.

A second limitation concerns the mismatch between high-performing deep models and the practical requirements of low-resource deployment. End-to-end fine-tuned CNNs can achieve strong performance when sufficient labelled data, GPU access, and training time are available, but their computational cost, storage footprint, and limited transparency can constrain use in edge-level or primary-care environments [22]. Moreover, post-hoc visual explanation methods such as Grad-CAM are often used to support clinical trust, yet saliency-based explanations may be unstable or clinically misleading when they highlight background regions or fail to reflect the true decision process [6], [7], [32]–[34]. This issue is particularly relevant for hybrid or frozen-backbone pipelines, where gradients may not correspond directly to the classifier that makes the final decision.

Handcrafted and deterministic morphology-based pipelines provide a contrasting paradigm. By explicitly encoding lesion count, size, shape, spatial distribution, confluence, and colour characteristics, such models can represent clinically meaningful aspects of rash morphology in a compact and inspectable feature space [8], [23]–[25]. This aligns with clinical visual reasoning, where lesion morphology and distribution patterns are central to exanthem assessment. However, morphology-only systems are vulnerable to segmentation errors caused by shadows, skin texture, uneven illumination, low image resolution, and ill-defined lesion boundaries. Consequently, purely handcrafted models may be auditable but brittle, whereas fully deep models may be robust but difficult to inspect.

This tension motivates a different framing for viral exanthem AI: the objective should not be to maximize closed-set accuracy alone, but to balance diagnostic discrimination, computational feasibility, feature-level auditability, and safe behaviour under uncertainty. Recent diagnostic-AI reporting guidance emphasizes that model evaluation should address not only performance metrics but also data provenance, intended use, applicability, generalizability, and deployment risks [NEW REF]. Similarly, early-stage clinical AI evaluation frameworks stress the need to consider safety, human factors, workflow integration, and the limits of algorithmic decision support before clinical deployment [NEW REF]. Within this perspective, a useful image-based triage model should be compact enough for constrained hardware, transparent enough for inspection, and cautious enough to avoid overconfident decisions outside its validated scope.

Against this background, the study is guided by three research questions. First, can explicit lesion morphology provide clinically meaningful discrimination between Varicella and Measles while remaining compact and auditable? Second, can frozen deep texture representations compensate for the segmentation fragility of deterministic morphology without requiring end-to-end CNN fine-tuning? Third, can morphology-informed fusion achieve a more favourable balance between diagnostic performance, computational cost, and decision-space transparency than morphology-only models, frozen transfer-learning baselines, or fully fine-tuned CNNs?

In this study, we propose MorphNN (FULL Form please), a morphology-informed frozen feature fusion framework for Varicella–Measles differentiation. MorphNN combines two complementary information streams. The first stream extracts eight deterministic morphology descriptors representing lesion burden, area distribution, circularity, spatial dispersion, confluence density, and localized colour characteristics. The second stream extracts high-level texture information from a frozen MobileNetV2 backbone and compresses the resulting embedding into ten principal components using training-set-only PCA. These features are concatenated into an 18-dimensional standardized representation and classified using a Random Forest ensemble. This design avoids end-to-end gradient adaptation while preserving a low-dimensional decision space that can be audited through feature-importance analysis. The study evaluates MorphNN through a four-phase experimental framework: morphology-only baselines, frozen transfer-learning baselines, end-to-end fine-tuned CNNs, and the proposed hybrid fusion model. This structure separates the contributions of explicit morphology, frozen deep texture representation, full CNN adaptation, and morphology-informed fusion. The evaluation reports macro F1-score, training time, inference latency, and deployable pipeline footprint under CPU-oriented benchmarking, reflecting the intended use case of resource-constrained triage rather than laboratory-only model comparison.

The main contributions of this work are as follows:

- We reframe Varicella–Measles image classification as a resource-constrained viral exanthem triage problem, where accuracy must be considered alongside auditability, computational efficiency, and uncertainty-aware use.
- We propose a compact morphology-informed frozen feature fusion framework that combines eight clinically interpretable lesion descriptors with ten PCA-compressed frozen CNN texture components.
- We provide a four-phase comparative evaluation against morphology-only classifiers, frozen transfer-learning models, and end-to-end fine-tuned CNN baselines, enabling a structured assessment of accuracy-efficiency trade-offs.
- We audit the final fused decision space using Random Forest feature importance, allowing explicit comparison between clinically meaningful morphology descriptors and latent texture components.
- [NEW RESULTS] - Add open-set or reject-option evaluation using Monkeypox, HFMD, Cowpox, and Healthy classes as unknown presentations. Report AUROC, AUPR, false-accept rate, rejection accuracy, or threshold-based referral behaviour.]
- [NEW RESULTS] - Add calibration and uncertainty analysis, including ECE, Brier score, reliability curves, and confidence-threshold behaviour for clinical triage.]

By positioning MorphNN as an auditable and resource-conscious triage framework rather than an accuracy-only classifier, this study aims to narrow the gap between high-performing dermatological image models and the practical requirements of low-resource infectious-disease screening. Accordingly, the present study should be interpreted as a deployment-oriented methodological evaluation, with clinical translation requiring subsequent open-set validation, calibration assessment, and prospective evaluation.

The remainder of this paper is organized as follows. Section 2 reviews

2. Related Work and Study Positioning:

2.1. Viral Exanthem Classification as a Triage Problem

Automated recognition of viral skin eruptions has gained renewed attention because several infectious conditions can produce visually overlapping cutaneous manifestations. Measles, Varicella, Monkeypox, HFMD, and related exanthems may differ substantially in transmission dynamics, isolation requirements, and clinical management, yet their image-level appearance can be difficult to separate when photographs are acquired under uncontrolled lighting, variable camera quality, incomplete clinical context, or atypical disease stage [10], [11], [13], [18], [19]. This makes viral exanthem recognition different from many conventional dermatological classification tasks: the goal is not only to assign a label, but to support triage decisions under uncertainty.

The Multi-Class Viral Skin Lesion Dataset (MCVSLD) provides a public image resource for studying viral and non-viral skin categories, including Chickenpox, Measles, Monkeypox, HFMD, Cowpox, and Healthy skin [13]. The availability of this dataset has enabled reproducible model comparison, but it also exposes an important methodological issue. Most image classifiers are trained and evaluated under closed-set assumptions, meaning that every test image is forced into one of the known training classes. This protocol is useful for benchmarking but does not fully represent clinical triage, where a submitted image may belong to a different rash condition, a non-infectious disorder, poor-quality acquisition, or an out-of-scope presentation. Therefore, performance evaluation for this domain should consider not only discrimination between known classes but also deployment constraints, feature auditability, and uncertainty-aware use.

[NEW REF] - Add an open-set recognition reference here, e.g., Bendale and Boulton/OpenMax or a recent visual open-set recognition survey. – search some other related one too, if you couldn't find it, let me know I will help you]

[NEW REF] - Add a reject-option/selective-classification reference here if the new experiments include referral or abstention thresholds.]

2.2. Deep Learning for Viral Lesion Classification

Convolutional neural networks pre-trained on large-scale image corpora have become the dominant approach for automated skin-lesion analysis. Architectures such as ResNet [15], EfficientNet [16], MobileNetV2 [17], VGG, SqueezeNet [REF], and transformer-based variants [REF] are widely used because transfer learning can extract discriminative visual representations from relatively small medical image datasets [14]. In viral lesion classification, this paradigm has been applied mainly to Monkeypox, Chickenpox, Measles, and related rash categories.

Nayak et al. [18] evaluated transfer-learning models, including ResNet and SqueezeNet, for Monkeypox, Measles, Chickenpox, and healthy skin classification, reporting encouraging performance while also using explainable AI tools to support interpretation. Savas [19] proposed a two-stage optimization strategy across a large set of pre-trained architectures for visually similar Monkeypox, Chickenpox, and Measles lesions, showing that architecture search and ensemble learning can improve classification performance. Abbas et al. [20] used transfer learning with explainable AI for skin disease prediction, while Alghoraibi et al. [21] integrated deep learning into a smart healthcare framework for

Mpox skin lesion monitoring. Collectively, these studies demonstrate the feasibility of CNN-based viral lesion classification and establish fine-tuned deep networks as strong accuracy-oriented baselines.

However, several limitations remain relevant for resource-constrained clinical triage. First, end-to-end fine-tuning requires computational resources that may not be available in primary-care or low-resource settings. Second, the resulting models often have larger storage and memory footprints than classical feature-based systems. Third, performance is usually reported under closed-set assumptions, with limited evaluation of uncertain or out-of-scope presentations. Fourth, interpretability is frequently provided through post-hoc visual saliency rather than through an auditable decision representation. These limitations do not invalidate deep learning approaches, but they motivate alternatives that explicitly consider computational feasibility, decision-space transparency, and deployment-oriented evaluation.

2.3. Lightweight Dermatological AI and Morphology-Based Feature Engineering

A contrasting line of work uses handcrafted or deterministic features to encode clinically meaningful image properties. In dermatology, morphology, colour, texture, and spatial distribution have long been central to visual assessment, and handcrafted descriptors have been used in lesion analysis for decades [23], [24]. Such methods are attractive for constrained environments because they produce compact feature vectors, require little or no GPU support, and allow direct inspection of the variables used by the classifier. Recent work has renewed interest in lightweight and interpretable dermatological pipelines. For example, Solak [25] proposed HCHS-Net, a multimodal handcrafted framework combining colour histograms, Haralick texture descriptors, Hu moments [26], and metadata for skin lesion classification. This type of approach highlights an important methodological advantage: the model can expose which measurable lesion characteristics contribute to a prediction, rather than relying entirely on latent deep representations.

For viral exanthems, morphology-based features are clinically relevant because Varicella and Measles differ in lesion form, distribution, and confluence patterns. Varicella often presents with lesions at different stages of evolution, whereas Measles is classically associated with a more diffuse maculopapular eruption [10], [NEW REF]. A morphology-aware model can therefore encode lesion burden, area variation, circularity, spatial dispersion, confluence density, and colour properties in a way that is more directly inspectable than an end-to-end CNN embedding.

The limitation is that deterministic morphology depends on the quality of preprocessing and segmentation. Shadows, skin pores, uneven illumination, low resolution, background clutter, and ill-defined lesion borders can corrupt binary masks and distort geometric measurements. This makes morphology-only systems auditable but potentially brittle. The central methodological question is therefore not whether morphology should replace deep learning, but whether explicit morphology can be fused with compact deep texture information to improve robustness while preserving an inspectable decision space.

[NEW REF] - Add a clinical reference for Varicella rash morphology, preferably CDC or another authoritative clinical source.]

2.4. Hybrid Feature Fusion in Medical Image Analysis

Hybrid fusion methods seek to combine complementary information sources rather than relying on a single representation. In multimodal machine learning, fusion can improve robustness when different feature streams capture non-redundant aspects of the underlying phenomenon [27]. In medical image analysis, this principle is often used to combine image features with clinical variables, handcrafted descriptors, or multiple neural representations. Within dermatological AI, hybrid systems have been used to integrate CNN features with structured clinical information, texture representations, or transformer-based attention mechanisms. Wang et al. [28] combined imaging and clinical features for skin lesion risk prediction, using explanation tools to support interpretation. Nie et al. [29] proposed a CNN-transformer hybrid architecture to combine local convolutional patterns with global attention for dermoscopic image classification. Gulzar et al. [30] demonstrated that lightweight CNN design choices,

including depthwise operations and squeeze-and-excitation modules, can reduce computational burden while maintaining competitive classification performance.

These studies support the broader value of combining representation types, but most do not directly address the specific Varicella–Measles triage setting. More importantly, many hybrid systems still rely on trainable deep networks, post-hoc saliency, or relatively high-dimensional latent features. In contrast, the present study evaluates whether a compact fusion space consisting of eight deterministic morphology descriptors and ten PCA-compressed frozen texture components can approach fine-tuned CNN performance while preserving a feature-level audit pathway. This shifts the emphasis from architecture complexity to a more deployment-relevant balance among accuracy, computational cost, and decision-space transparency.

2.5. Interpretability, Auditability, and the Limits of Post-Hoc Saliency

Interpretability is central to clinical AI because diagnostic support systems must be inspected, challenged, and understood by human users before they can be trusted in practice. Rudin [31] argues that high-stakes domains should prioritize interpretable models where possible rather than relying solely on explanations of black-box systems. In dermatological imaging, however, many studies still depend on post-hoc visual explanation methods such as Grad-CAM [6], [32]. Grad-CAM and related saliency approaches can be useful for visualizing class-discriminative regions, but their reliability has been questioned. Adebayo et al. [33] showed that some saliency methods may produce visually plausible maps even when model parameters or labels are randomized. Suara et al. [7], [34] further highlighted concerns in medical imaging, where heatmaps may emphasize clinically irrelevant regions or fail to correspond to expert-meaningful evidence. These concerns are especially important in frozen-backbone or hybrid pipelines: when the final classifier is not the same differentiable head used to compute gradients, saliency maps may not faithfully explain the actual decision mechanism.

Model-agnostic explanation methods such as SHAP [35] and LIME [36] provide alternatives for feature attribution, and tree-based models can expose global feature-importance patterns directly over structured input variables. Nevertheless, feature importance should be interpreted carefully. Importance over explicit lesion descriptors, such as confluence density or lesion area, has a clearer clinical meaning than importance over latent PCA-compressed CNN texture components. Therefore, the appropriate claim for a hybrid model is not complete clinical interpretability, but feature-level auditability of the final decision space. **This distinction is central to the present study. MorphNN does not claim that the frozen CNN texture stream is fully interpretable at the image-region level. Instead, it provides an auditable fused representation in which clinically meaningful morphology features and latent texture components can be inspected jointly.** This allows the analysis to determine whether the classifier depends primarily on explicit lesion structure, latent texture information, or a combination of both.

[NEW REF - Add a calibration reference, e.g., Guo et al. on neural network calibration, if calibration experiments are included., or find some other related one – again if help needed let me know]

2.6. The proposed approach - MorphNN

The literature reviewed above suggests three unresolved needs in viral exanthem AI. First, current viral lesion classifiers often emphasize closed-set accuracy but less frequently evaluate computational feasibility under constrained hardware conditions. Second, interpretability is commonly addressed through post-hoc saliency rather than through an auditable decision representation. Third, morphology-based descriptors are clinically meaningful and efficient but remain vulnerable to segmentation noise, suggesting the need for complementary texture information.

MorphNN is designed to address this intersection. It combines deterministic lesion morphology with frozen deep texture features in a compact 18-dimensional representation classified by a Random Forest ensemble. The morphology stream contributes explicit descriptors of lesion burden, geometry, spatial distribution, confluence, and localized colour characteristics. The frozen MobileNetV2 stream

contributes complementary texture information without end-to-end fine-tuning, and PCA compression restricts the latent feature contribution to a small number of components. This design directly supports the study's central premise: a viral exanthem triage model should not be evaluated by accuracy alone, but by the balance it achieves among discrimination, computational efficiency, deployability, and feature-level auditability.

Accordingly, the role of MorphNN is not to replace high-capacity CNNs as universal dermatological classifiers. Rather, it is positioned as a deployment-oriented methodological framework for a constrained and clinically relevant binary triage problem. The present evaluation compares morphology-only baselines, frozen transfer-learning models, fine-tuned CNNs, and hybrid fusion under a unified experimental protocol.

[NEW RESULTS - After completing the new experiments, add a sentence here summarizing how open-set rejection and calibration extend this positioning from closed-set classification toward safer triage.]. It is worth noting that the current study supports deployment-oriented benchmarking and feature-level audit of a Varicella–Measles classifier. However, translation to practice would require further validation under open-set conditions, calibration assessment, prospective evaluation, and testing across more diverse image sources and patient populations.

A few things to update after you run the new experiments:

1. Add a short **“Open-set and uncertainty-aware triage”** paragraph under Section 2.5 or 2.6, using the open-set/reject-option references marked [NEW REF].
2. Revise Table 1 so it does not claim MorphNN is simply “the only” method. A stronger column set would be: **closed-set task, open-set/reject evaluation, CPU-only benchmarking, no fine-tuning, deployable footprint, feature-level audit, calibration reported.**
3. Replace “clinical deployment” wording with **deployment-oriented evaluation** unless prospective clinical testing is added.

3. Materials and Methods:

3.1. Study Design

This study was designed as a deployment-oriented methodological evaluation of a morphology-informed image classification framework for Varicella–Measles differentiation. The primary objective was not only to compare classification performance, but also to examine whether an explicit, low-dimensional lesion representation could provide a more auditable and computationally efficient alternative to fully fine-tuned convolutional neural networks. The evaluation was therefore structured around four complementary questions: how far deterministic morphology alone can discriminate between the two conditions; how much diagnostic information is available in frozen deep texture representations; what performance ceiling is achieved by end-to-end CNN fine-tuning; and whether morphology-informed fusion improves the accuracy–efficiency–auditability trade-off.

The study followed a retrospective image-based experimental design using a public viral skin lesion dataset [REF of dataset]. Model development, cross-validation, dimensionality reduction, and classifier selection were performed only on the training partition. The holdout test partition was kept separate from model fitting and hyperparameter selection and was used for final generalization assessment. Reporting was organized to support transparent evaluation of the data source, preprocessing, model pipeline, baselines, benchmarking conditions, statistical analysis, and deployment-oriented constraints.

[NEW REF - Add STARD-AI here to show the diagnostic-AI reporting alignment.].

3.2. Study Design

The empirical evaluation used a binary subset of the Multi-Class Viral Skin Lesion Dataset (MCVSLD), accessed through the NIAID Data Ecosystem Discovery Portal and retrieved from

Mendeley Data [13]. The broader dataset contains images from multiple viral and non-viral skin categories, including Chickenpox, Measles, Monkeypox, Hand-Foot-and-Mouth Disease, Cowpox, and Healthy skin [13]. The present study focused on Varicella–Measles differentiation because these two conditions can present with visually overlapping exanthem patterns in routine photographs while requiring different clinical interpretation and infection-control responses.

For the closed-set binary experiment, images labelled as Chickenpox and Measles were retained. The current experimental split contains 900 Chickenpox and 660 Measles images in the training partition, and 113 Chickenpox and 83 Measles images in the holdout test partition. The holdout partition was not used during training, cross-validation, PCA fitting, standardization, or model selection.

[NEEDS CLARIFICATION - Verify and explicitly document whether the 113 Chickenpox and 83 Measles holdout images are provided by the dataset as a separate test split, derived from the public dataset split, or obtained from an additional source..]. Table 1 summarizes the binary closed-set dataset used in the present study.

Table 1: Closed-set Varicella–Measles dataset split used for model development and holdout evaluation.

Class	Training images	Holdout images	Total images used
Chickenpox / Varicella	900	113	1,013
Measles	660	83	743
Total	1,560	196	1,756

[NEW RESULTS - If the open-set experiment is added, include a second dataset table here listing Monkeypox, HFMD, Cowpox, and Healthy images used only as unknown or out-of-distribution presentations.]

3.3. Image Preprocessing and Augmentation

All preprocessing operations were applied in a partition-aware manner to avoid information leakage. Augmentation was applied only to the training partition and not to the holdout test partition. The augmentation protocol consisted of random horizontal flipping, discrete rotation, and centre cropping. This increased the effective training pool to 2,700 Chickenpox samples and 1,980 Measles samples. The holdout images remained unaugmented to preserve an independent estimate of generalization under the original data distribution. For the morphology stream, images were processed using colour normalization, noise-preserving smoothing, local contrast enhancement, adaptive thresholding, and contour filtering. For the frozen texture stream, images were resized to the input resolution required by the CNN backbone and normalized using ImageNet channel statistics. All transformations used for PCA fitting and feature standardization were fitted on the training data only and then applied unchanged to validation and holdout samples.

3.4. Deterministic Morphological Feature Extraction

The morphology stream was designed to convert each image into a compact, scale-normalized representation of visible lesion structure. This stream maps an input RGB image into an eight-dimensional feature vector representing lesion burden, lesion size, size heterogeneity, shape regularity, spatial dispersion, confluence density, and localized colour characteristics. The intention was not to create a perfect clinical segmentation mask, but to generate reproducible structural descriptors that could be inspected and audited at the classifier level.

First, a Gray-World white-balance transformation was applied to reduce global colour cast caused by heterogeneous image acquisition. A bilateral filter was then used to suppress fine-scale image noise while preserving lesion boundaries. Local contrast was enhanced using contrast-limited adaptive histogram equalization, followed by adaptive thresholding to obtain a candidate lesion mask. Morphological opening with an elliptical structuring element was used to suppress isolated noise. Candidate contours were retained only if their area fell within predefined relative bounds, thereby excluding very small artifacts and near-full-frame regions unlikely to represent individual lesions.

The retained contours were used to compute eight deterministic descriptors. All area-based and distance-based quantities were normalized by image size to reduce dependence on camera resolution. Table 2 defines each feature and its intended role in the decision representation.

Table 2: Morphological descriptors used in the eight-dimensional deterministic feature vector.

Feature	Descriptor	Operational definition	Rationale within the model
f1	Lesion count	Number of retained lesion-like contours	Captures visible lesion burden and fragmentation of the candidate lesion field
f2	Average lesion area	Mean contour area normalized by total image area	Represents typical lesion footprint while reducing resolution dependence
f3	Area heterogeneity	Standard deviation of contour areas normalized by total image area	Encodes variability in lesion size across the image
f4	Average circularity	Mean isoperimetric circularity of retained contours	Distinguishes compact rounded lesions from irregular or confluent regions
f5	Sparsity score	Mean nearest-neighbour centroid distance normalized by maximum image dimension	Captures spatial dispersion or clustering of lesions
f6	Confluence density	Total retained lesion area divided by total image area	Represents aggregate lesion coverage and coalescence
f7	Localized hue	Mean hue within the retained lesion mask	Captures localized colour tendency after normalization
f8	Localized saturation	Mean saturation within the retained lesion mask	Captures colour intensity within candidate lesion regions
Feature	Descriptor	Operational definition	Rationale within the model

These features provide an explicit decision interface but remain dependent on the quality of preprocessing and segmentation. This limitation motivated the addition of a complementary frozen texture stream.

3.5. Frozen Texture Embedding and PCA Compression

The second stream was designed to capture texture information not reliably represented by deterministic morphology. A MobileNetV2 backbone pretrained on ImageNet was used as a fixed feature extractor [14], [17]. The classification layer was removed, and global average pooling was applied to obtain a 1280-dimensional image-level embedding. The backbone weights were frozen throughout this phase, meaning that no end-to-end gradient adaptation was performed on the viral lesion images.

To reduce the dimensionality and deployment burden of the frozen embedding, principal component analysis was fitted on the training-set embeddings only. The leading ten principal components were retained, producing a compact texture vector for each image. The PCA dimensionality was selected empirically in the original experiment because classification performance plateaued at ten components, while retaining substantially more components increased dimensionality without improving macro F1. The resulting ten-dimensional vector was treated as a latent texture summary rather than a clinically interpretable image-region explanation.

[NEW RESULTS] - Add a PCA sensitivity table or curve showing macro F1 across component counts, e.g., 5, 10, 20, 50, and 95%-variance components. Write and highlight whether ten components remain optimal after the revised experiments.]

3.6. MorphNN Fusion and Classification

MorphNN combines the eight deterministic morphology descriptors and the ten PCA-compressed frozen texture components into a single 18-dimensional feature vector. Before classification, the concatenated vector is standardized using statistics estimated from the training data only. The standardized representation is then passed to a Random Forest classifier with balanced class weighting. The Random Forest was selected as the primary fusion classifier because it can model non-linear interactions among low-dimensional features while preserving a feature-level audit mechanism. In this setting, auditability refers to inspection of the final decision space through feature-importance analysis. It does not imply that every latent CNN texture component is clinically interpretable at the image-

region level. The morphology features have direct operational meaning, whereas the PCA-compressed CNN features represent latent texture variation. Table 3 summarizes the information flow through the MorphNN pipeline.

Table 3: MorphNN pipeline components

Pipeline component	Input	Output	Trainable in MorphNN?	Primary role
Morphological extractor	RGB clinical image	Eight deterministic morphology features	No	Provides explicit lesion-structure descriptors
Frozen MobileNetV2 backbone	Resized and normalized RGB image	1280-dimensional texture embedding	No	Extracts high-level texture representation
PCA projection	1280-dimensional embedding	Ten-dimensional texture vector	Fitted on training data only	Reduces texture dimensionality and footprint
Standardization	18-dimensional concatenated vector	Standardized 18-dimensional vector	Fitted on training data only	Aligns feature scales before classification
Random Forest classifier	Standardized fusion vector	Predicted class and class probability	Yes	Performs final decision and enables feature-level audit

3.7. Comparative Experimental Design

The evaluation was organized into four phases to isolate the contribution of each representation strategy, as presented in Table 4. This structure allows MorphNN to be compared not only with individual baselines, but also against the major modelling paradigms used in the literature: handcrafted morphology, frozen transfer learning, end-to-end fine-tuning, and hybrid feature fusion.

Table 4: Four-phase experimental evaluation

Phase	Model family	Representation	Classifier / head	Purpose
Phase 1	Morphology-only classical baselines	Eight deterministic morphology features	Random Forest, SVM, KNN	Estimate the discriminative value and auditability of explicit lesion morphology alone
Phase 2	Frozen transfer-learning baselines	Global average pooled CNN embeddings from frozen backbones	Linear classification head	Measure the diagnostic value of generic frozen texture features without fine-tuning
Phase 3	End-to-end fine-tuned CNNs	Fully adapted CNN representations	CNN classification layer	Establish the performance ceiling of pure deep learning under domain-specific adaptation
Phase 4	MorphNN hybrid fusion	Eight morphology features + ten PCA-compressed frozen texture components	Random Forest and alternative classical classifiers	Evaluate whether compact fusion improves the accuracy–efficiency–auditability trade-off

The CNN backbones evaluated in Phases 2 and 3 included MobileNetV2, MobileNetV3, ShuffleNetV2, SqueezeNet, ResNet18, ResNet34, and EfficientNet-B0. Phase 4 used frozen feature vectors derived from these backbones and combined them with the morphology vector before classification. The primary MorphNN configuration was MobileNetV2 + Random Forest, selected based on the observed balance of macro F1-score, training time, inference latency, and deployable footprint.

3.8. Evaluation Metrics

Macro F1-score was used as the primary classification metric because the class distribution was moderately imbalanced and equal weighting of Varicella and Measles performance was required. Holdout error count and confusion matrices were used to describe the direction of misclassification. Deployment-oriented metrics included training time, per-image inference latency, and deployable pipeline footprint. Table 5 summarizes the evaluation endpoints.

Table 5: Evaluation endpoints for closed-set and deployment-oriented assessment.

Evaluation dimension	Metric	Purpose
Classification performance	Macro F1-score	Primary measure of balanced class discrimination
Generalization	Holdout macro F1 and error count	Evaluate performance on the independent test partition
Error structure	Confusion matrix	Identify whether errors are concentrated in one diagnostic direction
Training efficiency	Training time	Quantify model-development cost
Runtime efficiency	CPU inference latency per image	Assess suitability for constrained deployment
Storage efficiency	Deployable pipeline footprint	Estimate model size and deployment burden
Auditability	Feature-importance profile	Inspect contribution of morphology and latent texture features

[**NEW RESULTS** - Add AUROC, AUPR, false-accept rate, and rejection accuracy if open-set/OOD experiments are completed.]

[**NEW RESULTS** - Add ECE, Brier score, reliability curves, and threshold-based referral rate if calibration experiments are completed.]

3.9. Statistical Analysis and Benchmarking Protocol

Five-fold stratified cross-validation was performed on the training partition for model development and comparative assessment. The holdout partition was used only for final evaluation. Repeated benchmarking was conducted over 20 runs to estimate runtime variability and provide stable performance summaries. Pairwise Wilcoxon signed-rank tests were used to compare repeated macro F1 and efficiency measurements between model configurations [38]. All inference benchmarking was conducted under CPU-oriented ONNX Runtime conditions. To reduce hardware-induced variability, CPU power-management effects were controlled by disabling dynamic boosting features, locking the processor to its base frequency using the Linux performance governor, and binding benchmarking threads to isolated physical cores. The software environment was based on Manjaro Linux with KDE Plasma. Training and benchmarking were evaluated across two hardware configurations, summarized in Table 6.

Table 6: Hardware configurations used for development and benchmarking.

Hardware	CPU	GPU	RAM	Intended role
Laptop	Intel i5-12400H	None	24 GB	CPU-only inference benchmarking
Desktop	Intel i5-13400	RTX 4070, 12 GB VRAM	32 GB	Training support for fine-tuned CNN baselines

3.10. Open-Set Rejection, Calibration, and Uncertainty Analysis

To support the uncertainty-aware triage framing introduced in Section 1, we extend the closed-set Varicella–Measles evaluation with open-set rejection and calibration analysis. MorphNN will be trained only on Varicella and Measles images, while Monkeypox, Hand-Foot-and-Mouth Disease, Cowpox, and Healthy images from MCVSLD will be withheld from training and introduced during evaluation as unknown or out-of-scope presentations. This protocol assesses whether the model can separate validated target conditions from visually related non-target categories rather than forcing every input into the binary decision space.

Reject-option behaviour will be evaluated using uncertainty scores derived from Random Forest outputs, including maximum predicted class probability, prediction entropy, and vote-margin difference between the two target classes. Thresholds for accept/reject decisions will be selected using validation data only. Images above the acceptance threshold will receive a Varicella–Measles prediction, whereas uncertain or out-of-scope images will be assigned to a reject/refer category. Open-set performance will be reported using known-versus-unknown AUROC, AUPR, false-accept rate, rejection accuracy, accepted-case macro F1, and threshold-based referral behaviour.

[Calibration will be assessed on the Varicella–Measles holdout partition using Expected Calibration Error, Brier score, and reliability-curve analysis. Where post-hoc calibration is applied, calibration parameters will be fitted only on validation data and evaluated on the independent holdout set. This analysis will determine whether MorphNN’s predicted confidence is suitable for triage-oriented thresholding rather than only closed-set class assignment.]

4. Results

4.1. Overview of Results

This section presents the empirical evaluation of MorphNN across four experimental phases: morphology-only classification, frozen transfer learning, end-to-end CNN fine-tuning, and morphology-informed frozen feature fusion. The results are interpreted according to the central objective of the study: to determine whether a compact, auditable fusion representation can approach fine-tuned CNN performance while improving computational efficiency and preserving a feature-level audit pathway.

The results show a consistent evidence chain. First, explicit morphology provides useful diagnostic information but remains vulnerable to segmentation and acquisition artifacts. Second, frozen CNN texture features alone provide complementary visual information but do not achieve the strongest closed-set discrimination. Third, end-to-end fine-tuned CNNs establish the highest closed-set holdout ceiling, but at substantially greater training cost. Fourth, MorphNN reaches near-parity with the best fine-tuned CNN on the independent holdout set while providing a substantially more favourable accuracy–efficiency–auditability trade-off under CPU-oriented benchmarking. Table 7 summarizes the best-performing model from each experimental phase.

Table 7: Summary of best-performing models across the four experimental phases.

Phase	Best model	Representation	Repeated/CV macro F1	Holdout macro F1	Errors / 196	Training time (s)	Inference latency (ms/image)
Phase 1	Random Forest	Eight morphology features	0.855 ± 0.020	0.901	19	3.21 ± 0.18	0.31 ± 0.03
Phase 2	EfficientNet-B0 frozen	Frozen CNN embedding	0.896 ± 0.014	0.934	13	12.4 ± 0.7	12.1 ± 0.8
Phase 3	EfficientNet-B0 fine-tuned	End-to-end adapted CNN	0.941 ± 0.009	0.963	7	312 ± 14.3	12.8 ± 0.9
Phase 4	MorphNN MobileNetV2 + RF	Morphology + PCA-compressed frozen texture	0.953 ± 0.007	0.957	8	14.2 ± 0.6	9.8 ± 0.4

The independent holdout comparison shows that fine-tuned EfficientNet-B0 achieved the highest closed-set macro F1-score, while MorphNN achieved near-parity with only one additional error. Therefore, MorphNN provides a more deployment-oriented operating point by combining near fine-tuned CNN performance with reduced training time, smaller footprint, CPU-feasible inference, and feature-level auditability.

4.2. Morphology-Only Baseline

The first phase evaluated whether the eight deterministic morphology descriptors alone could discriminate between Varicella and Measles. Three classical classifiers were trained using the same morphology vector: Random Forest, KNN, and SVM. The results are shown in Table 8.

Table 8: Morphology-only classification using the eight-dimensional deterministic feature vector.

Classifier	CV macro F1	CV SD	Holdout macro F1	Training time (s)	Inference latency (ms/image)	Size (MB)	Errors / 196
Random Forest	0.855	0.020	0.901	3.21 ± 0.18	0.31 ± 0.03	2.4	19
KNN, k = 5	0.836	0.012	0.810	0.04 ± 0.01	0.19 ± 0.02	0.3	36
SVM, RBF	0.795	0.012	0.783	1.87 ± 0.11	0.24 ± 0.02	0.6	41

Random Forest achieved the best morphology-only result, with holdout macro F1 of 0.901 and 19 errors on the 196-image holdout partition. This confirms that explicit lesion morphology contains meaningful discriminative information. However, the residual error count indicates that morphology alone is insufficient for robust Varicella–Measles differentiation, particularly where segmentation is affected by shadows, skin texture, low image quality, or ill-defined lesion boundaries.

The morphology-only baseline nevertheless establishes the most computationally efficient operating point. Random Forest inference required only 0.31 ± 0.03 ms/image, indicating that explicit morphology is highly suitable for constrained computation but not sufficient as a standalone representation.

4.3. Frozen Transfer-Learning Baselines

The second phase evaluated pretrained CNN backbones as frozen texture extractors. This phase tested whether generic deep visual representations, without end-to-end adaptation, could provide sufficient discriminative information. Table 9 reports the results.

Table 9: Frozen transfer-learning baseline performance.

Backbone	CV macro F1	Training time (s)	Inference latency (ms/image)	Size (MB)
EfficientNet-B0	0.896 ± 0.014	12.4 ± 0.7	12.1 ± 0.8	20.4
ResNet34	0.891 ± 0.015	11.2 ± 0.6	12.4 ± 0.9	83.2
ResNet18	0.882 ± 0.016	9.7 ± 0.5	10.3 ± 0.7	44.6
MobileNetV2	0.874 ± 0.018	8.3 ± 0.4	9.2 ± 0.6	13.4
MobileNetV3	0.861 ± 0.021	7.9 ± 0.4	8.7 ± 0.5	20.9
ShuffleNetV2	0.843 ± 0.019	6.1 ± 0.3	5.8 ± 0.4	8.7
SqueezeNet	0.819 ± 0.023	5.4 ± 0.3	4.9 ± 0.4	4.7

EfficientNet-B0 achieved the strongest frozen-backbone result, with macro F1 of 0.896 ± 0.014 . However, frozen transfer learning did not exceed the holdout macro F1 of the morphology-only Random Forest baseline. This finding is important because it shows that generic CNN texture information is useful but incomplete when used without either explicit morphology or disease-specific adaptation. Frozen CNN inference also introduced a substantial runtime cost relative to morphology-only classification. EfficientNet-B0 required 12.1 ± 0.8 ms/image, compared with 0.31 ± 0.03 ms/image for morphology-only Random Forest. Thus, frozen texture features alone do not provide the optimal trade-off between discrimination and computational efficiency.

4.4. End-to-End Fine-Tuned CNNs

The third phase evaluated the upper closed-set performance achievable by pure deep learning. All seven CNN backbones were fine-tuned end to end on the training partition. Table 10 summarizes the results.

Table 10: Frozen transfer-learning baseline performance.

Backbone	CV macro F1	Training time (s)	Inference latency (ms/image)	Size (MB)
EfficientNet-B0	0.941 ± 0.009	312 ± 14.3	12.8 ± 0.9	20.5

ResNet34	0.934 ± 0.010	298 ± 13.1	13.1 ± 1.0	83.4
ResNet18	0.921 ± 0.011	231 ± 10.8	11.2 ± 0.8	44.8
MobileNetV2	0.912 ± 0.012	184 ± 9.2	10.1 ± 0.7	13.5
MobileNetV3	0.904 ± 0.015	176 ± 8.7	9.4 ± 0.7	21.0
ShuffleNetV2	0.883 ± 0.017	142 ± 7.4	6.5 ± 0.5	8.8
SqueezeNet	0.851 ± 0.020	118 ± 6.9	5.6 ± 0.4	4.8

Fine-tuning improved macro F1 across all backbones, confirming the value of disease-specific representation learning. EfficientNet-B0 achieved the strongest fine-tuned repeated evaluation result, with macro F1 of 0.941 ± 0.009 , and later achieved the highest independent holdout macro F1 of 0.963. The performance improvement came with a substantial training burden. Fine-tuned EfficientNet-B0 required 312 ± 14.3 s of training, while fine-tuned ResNet34 required 298 ± 13.1 s. These results establish fine-tuned CNNs as strong accuracy-oriented references, but also clarify the deployment gap addressed by MorphNN: maximum closed-set accuracy is achievable, but not necessarily with the most favourable efficiency, footprint, or auditability profile.

4.5. MorphNN Hybrid Fusion Performance

The fourth phase evaluated MorphNN by combining eight deterministic morphology features with ten PCA-compressed frozen CNN texture components. The resulting 18-dimensional standardized vector was classified using classical machine-learning models. Table 11 compares the principal MorphNN configurations against their corresponding fine-tuned CNN baselines.

Table 11: MorphNN hybrid fusion compared with fine-tuned CNN baselines

Backbone + classifier	MorphNN macro F1	Fine-tuned CNN macro F1	p-value	MorphNN training time (s)	Fine-tuned training time (s)	p-value	MorphNN inference (ms/image)	Fine-tuned inference (ms/image)	p-value
MobileNetV2 + RF	0.953 ± 0.007	0.912 ± 0.012	<0.001	14.2 ± 0.6	184 ± 9.2	<0.001	9.8 ± 0.4	10.1 ± 0.7	0.043
EfficientNet-B0 + RF	0.949 ± 0.008	0.941 ± 0.009	0.031	18.1 ± 0.8	312 ± 14.3	<0.001	10.6 ± 0.5	12.8 ± 0.9	0.038
ResNet18 + RF	0.946 ± 0.009	0.921 ± 0.011	0.024	16.3 ± 0.7	231 ± 10.8	<0.001	10.1 ± 0.5	11.2 ± 0.8	0.041
ResNet34 + RF	0.945 ± 0.009	0.934 ± 0.010	0.038	17.4 ± 0.7	298 ± 13.1	<0.001	10.9 ± 0.5	13.1 ± 1.0	0.036
MobileNetV2 + SVM	0.941 ± 0.010	0.912 ± 0.012	0.044	13.8 ± 0.6	184 ± 9.2	<0.001	9.6 ± 0.4	10.1 ± 0.7	0.047
MobileNetV3 + RF	0.938 ± 0.011	0.904 ± 0.015	0.029	13.1 ± 0.6	176 ± 8.7	<0.001	9.1 ± 0.4	9.4 ± 0.7	0.049

The best MorphNN configuration was MobileNetV2 + Random Forest, achieving macro F1 of 0.953 ± 0.007 , training time of 14.2 ± 0.6 s, and CPU inference latency of 9.8 ± 0.4 ms/image. The deployable pipeline footprint was 16.1 MB. Compared with fine-tuned MobileNetV2, MorphNN achieved higher macro F1 while reducing training time from 184 ± 9.2 s to 14.2 ± 0.6 s. Compared with fine-tuned EfficientNet-B0, MorphNN achieved substantially lower training time and a smaller deployable footprint while remaining close to the best holdout performance. These findings support the design hypothesis that explicit morphology and frozen texture information provide complementary signals. MorphNN does not rely on end-to-end gradient adaptation, yet it recovers much of the discriminative performance associated with fine-tuned CNNs.

4.6. Independent Holdout Evaluation

The independent holdout partition provides the most important generalization comparison because it was not used for training, cross-validation, PCA fitting, standardization, or model selection. Table 12 compares the best-performing model from each experimental phase.

Table 12: Holdout generalization across the four experimental phases

Phase	Best model	Holdout macro F1	Errors / 196	Phase
Phase 1	Random Forest, morphology only	0.901	19	Phase 1
Phase 2	EfficientNet-B0, frozen transfer	0.934	13	Phase 2
Phase 3	EfficientNet-B0, fine-tuned	0.963	7	Phase 3
Phase 4	MorphNN, MobileNetV2 + RF	0.957	8	Phase 4

The holdout hierarchy is central to the interpretation of the study. Morphology-only Random Forest achieved 0.901 macro F1, frozen EfficientNet-B0 improved this to 0.934, and fine-tuned EfficientNet-B0 achieved the highest result at 0.963. MorphNN achieved 0.957, corresponding to one additional error relative to fine-tuned EfficientNet-B0.

The results should therefore be interpreted as indicating near-parity rather than absolute superiority. MorphNN does not surpass fine-tuned CNN models as the highest-performing classifier under closed-set evaluation conditions. Instead, its primary contribution lies in achieving performance levels that closely approach the holdout accuracy ceiling of fine-tuned CNNs while simultaneously providing advantages in training efficiency, reduced deployment footprint, CPU-compatible inference, and enhanced model transparency through a feature-level audit mechanism.

[NEW EXPERIMENT] - Holdout confidence intervals and paired testing: compute bootstrap 95% confidence intervals for holdout macro F1 for the best model from each phase. Perform McNemar’s test or an equivalent paired test comparing MorphNN MobileNetV2 + RF with fine-tuned EfficientNet-B0 on the same holdout images. Insert the results after Table 12 and interpret whether the 0.957 versus 0.963 difference is statistically meaningful.]

4.7. Ablation and Representation Complementarity

The ablation analysis tested whether the MorphNN gain resulted from genuine complementarity between explicit morphology and frozen texture information. Table 13 summarizes the representation-level comparison.

Table 13: Ablation analysis of morphology, frozen texture, and fused representation

Representation	Model	Macro F1	Main interpretation
Morphology only	Random Forest	0.901	Clinically meaningful and efficient, but vulnerable to segmentation artifacts
Frozen texture only	MobileNetV2	0.874	Captures texture information but lacks explicit lesion-structure encoding
Morphology + frozen texture	MorphNN, MobileNetV2 + RF	0.953	Combines structural descriptors and complementary texture cues

This ablation result is one of the strongest pieces of evidence in the study. Neither stream is sufficient on its own: morphology-only classification reaches 0.901, and frozen MobileNetV2 texture alone reaches 0.874. The fused representation increases macro F1 to 0.953, indicating that the two streams contribute non-redundant information.

The error-mitigation matrix provides further support for this interpretation. Relative to the morphology-only baseline, MorphNN reclassified 7 Chickenpox cases and 4 Measles cases that had previously been misclassified. This suggests that the frozen texture stream helps correct failure modes arising from segmentation fragility, while the morphology stream continues to provide explicit lesion-structure information.

4.8. Feature-Level Audit

Feature-level audit was conducted using Random Forest feature importance over the fused 18-dimensional representation, as presented in Table 14. In the morphology-only model, confluence ratio and average lesion area were the dominant contributors. In the MorphNN model, the highest-

ranked features were CNN Texture 1 and confluence ratio, followed by additional texture components and morphology descriptors.

Model	Dominant features	Training time (s)	Inference latency (ms/image)
Morphology-only Random Forest	Confluence ratio, average lesion area	3.21 $\pm$ 0.18	0.31 $\pm$ 0.03
MorphNN, MobileNetV2 + RF	CNN Texture 1, confluence ratio	14.2 $\pm$ 0.60	9.80 $\pm$ 0.40

This result supports feature-level auditability while also clarifying the limits of the interpretability claim. The confluence ratio and average lesion area represent explicitly defined morphological descriptors with direct biological and clinical meaning. In contrast, CNN Texture 1 is a PCA-compressed latent representation and therefore should not be interpreted as clinically meaningful in the same manner. Accordingly, MorphNN is best characterised as auditable at the fused-feature level rather than fully interpretable at the image-region level.

The continued dominance of the confluence ratio in both the morphology-only and hybrid models further indicates that MorphNN retains explicit lesion morphology after the inclusion of CNN-derived texture features. Rather than replacing handcrafted morphology with latent representations, the hybrid model integrates CNN texture information alongside a clinically meaningful structural descriptor

[NEW EXPERIMENT] - Feature-audit validation: compute permutation importance for all 18 fused features on the holdout set. If feasible, also compute SHAP values for the Random Forest classifier. Report whether confluence ratio remains among the most important explicit morphology features and whether the ranking of CNN texture components is stable. Add a table comparing impurity importance, permutation importance, and SHAP importance.]

4.9. PCA Dimensionality Sensitivity

The current MorphNN configuration incorporates ten PCA-compressed frozen texture components. The reported performance suggests that this setting provides a compact yet effective representation. However, because retaining sufficient components to explain 95% of the cumulative variance would require substantially higher dimensionality, this dimensionality choice permits explicit evaluation. Such analysis is important because the compactness of the fused representation is a central element of the proposed contribution.

[NEW EXPERIMENT] - PCA sensitivity analysis: rerun MorphNN using different numbers of PCA texture components, for example 5, 10, 20, 50, and the number required for 95% cumulative variance. For each setting, report macro F1, training time, inference latency, and pipeline footprint. Confirm whether ten components remains the best accuracy–efficiency compromise.]

Suggested table:

Table 14: PCA dimensionality sensitivity for MorphNN

PCA components	Variance retained	Macro F1	Holdout macro F1	Training time (s)	Inference latency (ms/image)	Pipeline size
5	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
10	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
20	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
50	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
95% variance	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

4.10. Robustness Under Image Perturbations

The study is motivated by the challenge of non-standard clinical photographs, including variation in illumination, sharpness, compression, and framing. The current closed-set results demonstrate performance under the original dataset conditions, but an additional robustness analysis is needed to test whether MorphNN preserves its advantage when image quality is degraded.

[NEW EXPERIMENT] - Robustness testing: evaluate morphology-only RF, fine-tuned EfficientNet-B0, and MorphNN under controlled perturbations applied only at test time. Include brightness shift, contrast shift, Gaussian blur, JPEG compression, and resizing/downsampling. Report macro F1 degradation relative to the clean holdout set. Interpret whether MorphNN is more stable than morphology-only classification and whether it remains competitive with fine-tuned EfficientNet-B0.]

Suggested table:

Table 15: Robustness under controlled test-time perturbations

Model	Clean macro F1	Brightness shift	Contrast shift	Blur	JPEG compression	Downsampling	Mean performance drop
Morphology-only RF	0.901	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Fine-tuned EfficientNet-B0	0.963	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
MorphNN, MobileNetV2 + RF	0.957	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

4.11. Dataset Leakage and Duplicate Audit

Because the source dataset consists of publicly collected images, the reliability of holdout evaluation depends on excluding duplicate or near-duplicate leakage between training and test partitions. This audit is important for strengthening the credibility of the reported generalization results. (please link it with the new experiment)

[NEW EXPERIMENT] - Leakage and duplicate assessment: compute perceptual hashes or near-duplicate similarity scores across all training and holdout images. Report the number of exact duplicates, near-duplicates, and any removed or retained cases. Confirm that augmentation was applied only to the training partition and that no augmented image entered the holdout set.]

Suggested table:

Table 11..

Table 16: Dataset duplicate and leakage audit

Audit step	Method	Number detected	Action taken
Exact duplicate check	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Near-duplicate check	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Train–holdout overlap	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Augmentation leakage check	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

4.12. Deployment-Oriented Efficiency and Runtime Profile

The operational value of MorphNN is best interpreted by jointly considering classification performance, training cost, inference latency, deployable footprint, and CPU runtime behaviour. Morphology-only Random Forest provides the lowest-latency configuration, achieving 0.31 ± 0.03 ms/image inference with a 2.4 MB footprint, but its holdout macro F1-score remains lower at 0.901. Fine-tuned EfficientNet-B0 provides the strongest closed-set holdout performance, reaching 0.963 macro F1, but requires substantially greater training time and a larger deployable pipeline. MorphNN occupies a more balanced operating point. The MobileNetV2 + Random Forest configuration achieved 0.957 holdout macro F1, closely approaching the fine-tuned EfficientNet-B0 result, while requiring 14.2 ± 0.6 s of training, 9.8 ± 0.4 ms/image CPU inference, and a 16.1 MB deployable footprint. Compared with fine-tuned EfficientNet-B0, this corresponds to approximately a 22-fold reduction in training time and a 21.5% reduction in deployable footprint, with only one additional holdout error.

The CPU-frequency analysis further supports this deployment-oriented interpretation. The morphology-only Random Forest remained in the low-frequency execution band because it relies primarily on lightweight feature computation and tree traversal. Fine-tuned CNN models required sustained higher-frequency execution during convolutional inference. MorphNN showed an intermediate multi-stage profile: morphology extraction and Random Forest classification remained lightweight, while the frozen MobileNetV2 feature-extraction stage accounted for the main computational load. This explains why MorphNN is slower than morphology-only classification but remains CPU-feasible while providing substantially stronger diagnostic performance.

These results show that MorphNN is not merely a competitive classifier in terms of macro F1-score. Its main contribution is the operating point it establishes: near fine-tuned CNN holdout performance with lower training cost, smaller deployment footprint, sub-10 ms CPU inference, and feature-level auditability through a compact 18-dimensional decision representation.

4.13. Open-Set Rejection and Threshold-Based Triage

The closed-set evaluation shows that MorphNN can approach fine-tuned CNN performance when all test images belong to either Varicella or Measles. However, real triage requires the model to handle images outside the validated binary decision space. Open-set rejection is therefore required to test whether the model can avoid unsafe forced classification when exposed to visually related but non-target presentations.

[NEW EXPERIMENT - Open-set rejection: train MorphNN only on Varicella and Measles. Withhold Monkeypox, HFMD, Cowpox, and Healthy images from training and introduce them only during evaluation as unknown or out-of-scope presentations. Evaluate morphology-only RF, fine-tuned EfficientNet-B0, and MorphNN using known-versus-unknown AUROC, AUPR, false-accept rate, rejection accuracy, accepted-case macro F1, and referral rate. Thresholds must be selected using validation data only.]

Suggested table:

Table 17: Open-set rejection and threshold-based triage performance.

Model	Unknown classes	AUROC	AUPR	False-accept rate	Rejection accuracy	Accepted-case macro F1	Referral rate
Morphology-only RF	Monkeypox, HFMD, Cowpox, Healthy	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Fine-tuned EfficientNet-B0	Monkeypox, HFMD, Cowpox, Healthy	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
MorphNN, MobileNetV2 + RF	Monkeypox, HFMD, Cowpox,	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

	Healthy						
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[NEW EXPERIMENT] - Add threshold analysis: report how accepted-case macro F1 and referral rate change across multiple confidence thresholds. Include a curve or table showing the trade-off between coverage and error rate.]

Suggested table:

Table 18: Threshold-based referral behaviour for MorphNN

Acceptance threshold	Accepted coverage	Referral rate	Accepted-case macro F1	False-accept rate for unknowns	Error rate among accepted cases
[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

4.14. Calibration and Confidence Reliability

Because the revised study positions MorphNN as a triage-oriented framework, predicted confidence must be evaluated rather than assumed reliable. Calibration analysis is therefore required to determine whether predicted probabilities correspond to empirical correctness and whether confidence thresholds are meaningful for referral decisions.

[NEW EXPERIMENT] - Calibration analysis: compute Expected Calibration Error, Brier score, and reliability curves for morphology-only RF, fine-tuned EfficientNet-B0, and MorphNN on the Varicella–Measles holdout partition. If post-hoc calibration is used, fit calibration parameters on validation data only and evaluate calibrated model on the independent holdout set.]

Suggested table:

TableXXX. Calibration performance on the Varicella–Measles holdout partition.

Model	Calibration method	ECE	Brier score	Reliability-curve summary
Morphology-only RF	None / post-hoc calibrated	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
Fine-tuned EfficientNet-B0	None / post-hoc calibrated	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]
MorphNN, MobileNetV2 + RF	None / post-hoc calibrated	[NEW RESULTS]	[NEW RESULTS]	[NEW RESULTS]

[NEW EXPERIMENT] - Report how calibration affects threshold-based referral. Compare pre-calibration and post-calibration referral rate, accepted-case macro F1, false-accept rate for unknown images, and error rate among accepted cases.

This analysis will determine whether MorphNN’s predicted confidence is suitable for triage-oriented thresholding. A model may achieve strong macro F1 but still be poorly calibrated; therefore, calibration should be interpreted as a separate property from classification accuracy.]

4.15. Summary of Findings (please update it accordingly)

The closed-set results support four findings. First, explicit morphology provides a useful and highly efficient baseline but does not fully resolve Varicella–Measles ambiguity. Second, frozen CNN texture features alone are not sufficient, but they provide complementary information when

fused with morphology. Third, fine-tuned EfficientNet-B0 remains the strongest closed-set holdout classifier, achieving 0.963 macro F1. Fourth, MorphNN achieves near holdout parity at 0.957 macro F1 while reducing training time, reducing deployable footprint, maintaining sub-10 ms CPU inference, and enabling feature-level audit through an 18-dimensional representation. The additional experiments specified above will determine whether this favourable closed-set operating point extends to uncertainty-aware triage. In the final version, the open-set rejection, threshold-based referral, calibration, robustness, feature-audit validation, PCA sensitivity, and leakage-audit results should be integrated into the main results narrative rather than treated as supplementary checks (please adjust it accordingly based on the new results).

Abdullah for figures please – see below which of them are feasible, doable and can best fit

(For now if possible just do this one, then I will comeback again to you once you have done the above and this below figure.)

Fig. 1/2. MorphNN architecture and feature-fusion pipeline

Current source: not clearly strong enough as a figure in the manuscript; currently mostly algorithmic text.

Where to fit: Section 3.6 MorphNN Fusion and Classification

make this a major new schematic. This should be one of the most important figures in the paper.

[NEW FIGURE - Create a full MorphNN workflow diagram.]

Panel A: input clinical image.

Panel B: morphology stream: preprocessing → lesion mask → eight morphology descriptors.

Panel C: frozen texture stream: MobileNetV2 frozen backbone → 1280-dimensional embedding → PCA to 10 components.

Panel D: fusion: $8 + 10 = 18$ standardized features.

Panel E: Random Forest classifier → prediction + feature-level audit.

Caption: Fig. 2. MorphNN morphology-informed frozen feature fusion pipeline.

The framework combines eight deterministic lesion morphology descriptors with ten PCA-compressed frozen MobileNetV2 texture components. The fused 18-dimensional standardized vector is classified using a Random Forest ensemble, enabling feature-level audit of the final decision space without end-to-end CNN fine-tuning.